EE 102: Signal Processing and Linear Systems

Problem Set #9: Fourier Transform Properties and RC Applications

Name: _____ **Submission Date:** _____

Released October 26, 2026. Due **Wednesday, November 4, at 11:59 PM** (Pacific time). This shorter exam-week assignment has two written problems and is worth **100 points**. No programming is required.

Use the continuous-time Fourier transform convention

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega.$$

Frequencies ω are in rad/s. Show your reasoning; exact expressions are sufficient.

Problem 1 The RC circuit in the frequency domain. [50 points]

A series RC circuit has $R = 1\,\text{k}\Omega$ and $C = 1000\,\mu\text{F}$. The input $x(t)$ is the voltage across the series combination; the output $y(t)$ is the capacitor voltage. With zero initial stored energy, its impulse response is

$$h_C(t) = \frac{1}{\tau_C}e^{-t/\tau_C}u(t), \qquad \tau_C = RC.$$

(a) [15 points] Calculate τ_C. Derive $H_C(\omega)$ directly from the Fourier transform integral, and find $H_C(0)$.

(b) [15 points] Find the magnitude and phase of $H_C(\omega)$ at $\omega = 1$ and $\omega = 3$ rad/s. What happens to the magnitude as $|\omega| \to \infty$?

(c) [20 points] The input is

$$x(t) = 2\cos(t) + 2\cos(3t) \quad \text{volts},$$

present for all time. Find the bounded steady-state output voltage. Which input component is reduced more, and how does that relate to the name "low-pass filter"?

Hint: Use the LTI complex-exponential eigenfunction property from Weeks 6–7. A component $A\cos(\omega t)$ produces $A|H_C(\omega)|\cos(\omega t + \angle H_C(\omega))$ in steady state.

Problem 2 Symmetry and time shifting of a pulse. [50 points]

Define the aperiodic pulse

$$p(t) = \begin{cases} 2, & -1 < t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) [15 points] Sketch $p(t)$. Before calculating its transform, use its symmetry to predict whether $P(\omega)$ is real and whether it is even. Then derive $P(\omega)$ and give its value at $\omega = 0$.

Hint: For a real, even signal, $P(\omega) = 2\int_0^\infty p(t)\cos(\omega t)\,dt$. At zero frequency, use the signed area of the pulse.

(b) [20 points] Let $q(t) = p(t-2)$. Sketch $q(t)$ and obtain $Q(\omega)$ using the time-shifting property. Explain why $|Q(\omega)| = |P(\omega)|$ even though $q(t)$ is not even about $t = 0$. Find the magnitude and phase of $Q(\pi/4)$.

Hint: $x(t-t_0) \longleftrightarrow e^{-j\omega t_0}X(\omega)$, $\sin(\pi/4) = \sqrt{2}/2$, and $e^{-j\pi/2} = -j$.

(c) [15 points] Now let $r(t) = p(t-2) + p(t+2)$. Without performing another transform integral, find $R(\omega)$ and write it as a real, even expression. Evaluate $R(0)$ and $R(\pi/4)$. Explain how the two shifted pulses can cancel at one frequency even though both pulses are positive in time.

Hint: Apply linearity and time shifting, then use $e^{j\theta} + e^{-j\theta} = 2\cos\theta$.